1.
```
CREATE TABLE RICOVERO(
        IdRic           INT PRIMARY KEY,
        CFPaz           CHAR(16) NOT NULL,
        DataIngresso    DATE NOT NULL,
        DataDimissione  DATE CHECK (DataDimissione IS NULL OR DataDimissione >= DataIngresso),
        Reparto         VARCHAR(32) NOT NULL,
        Diagnosi        VARCHAR(64),
        FOREIGN KEY (CFPaz) REFERENCES PAZIENTE (CF) ON UPDATE CASCADE DELETE NO ACTION
)
```

2.
```
ALTER TABLE PAZIENTE ADD NumeroRicoveri Integer NOT NULL CHECK(NumeroRicoveri > 0)
```

3.
```
DELETE FROM PAZIENTE
WHERE CF NOT IN (
        SELECT CFPaz
        FROM RICOVERO
        WHERE DataIngresso >= “2000-01-01”
) AND CF IN (
        SELECT CFPaz
        FROM RICOVERO
        WHERE DataIngresso < “2000-01-01”
)
```

4.
```
SELECT RegioneRes, AVG(DataDimissione-DataIngresso)   -- la funzione corretta sarebbe DATEDIFF([day,]DataDimissione, DataIngresso)
FROM Paziente, Ricovero
WHERE CF = CFPaz AND YEAR(DataIngresso) = 2024 AND DataDimissione IS NOT NULL
GROUP BY RegioneRes
```

5.
```
SELECT CF, Nome, Cognome          -- NB: un paziente può avere un solo ricovero alla volta senza DataDimissione (= non concluso)
FROM PAZIENTE JOIN RICOVERO R ON CF = CFPaz
WHERE DataDimissione IS NULL
AND NOT EXISTS (
        SELECT *
        FROM RICOVERO R2 JOIN INTERVENTO I ON R2.IDRic=I.IDRic
        WHERE R.CFPaz = R2.CFPaz AND R.IDRic <>R2.IDRic
)

-- oppure

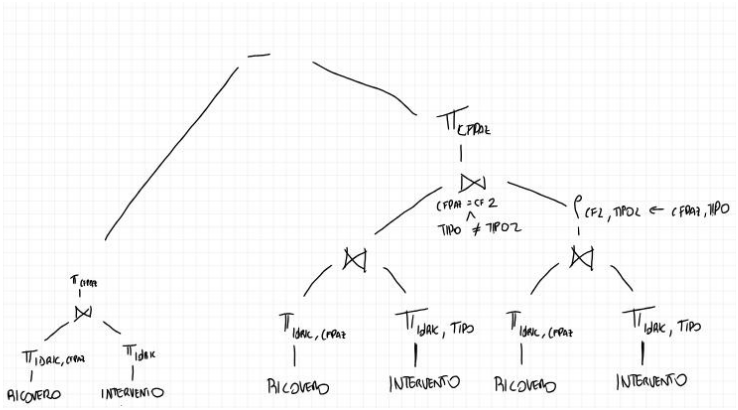
SELECT CF, Nome, Cognome
FROM PAZIENTE JOIN RICOVERO R ON CF = CFPaz
WHERE DataDimissione IS NULL
AND CF NOT IN(
        SELECT CFPaz
        FROM RICOVERO R2 JOIN INTERVENTO I ON R2.IDRic=I.IDRic
        WHERE DataDimissione IS NOT NULL
```

6.

```
CREATE VIEW NumPazAttRicRep(Reparto, Num) AS
        SELECT Reparto, COUNT(*)
        FROM Ricovero
        WHERE DataDimissione IS NULL
        GROUP BY Reparto
        HAVING COUT(*) >= 0.1 * (
                SELECT COUNT(*)
                FROM Ricovero
                WHERE DataDimissione IS NULL
        );

SELECT AVG(Num)
FROM NumPazAttRicRep;
```

LINGUAGGI FORMALI




```
Paz2TipiDiv(CF):-   INTERVENTO(_, IdR1, _, _, T1, _, _),
                    INTERVENTO(_, IdR2, _, _, T2, _, _),
                    RICOVERO(IdR1, CF, _, _, _, _),
                    RICOVERO(IdR2, CF, _, _, _, ),
                    T1 != T2

PazOK(CF):-         INTERVENTO(_, IdR, _, _, _, _, _),
                    RICOVERO(IdR, CF, _, _, _, _),
                    ¬Paz2TipiDiv(CF)

?- PazOK(x)
```

NB: attenzione a NON aggiungere IdR1 = IdR2 o IdR1 != IdR2